Rough or Just Noisy?

How Trade Data Change Volatility Estimates
and Proposed Investments

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Abstract

This paper asks whether different representations of the same recorded trades change a volatility estimate and the investment amount proposed from it. The comparison covers 1,990 fund-days for five exchange-traded funds over the same historical calendar. For SPY, the two sources produce materially different one-minute realized variance on 41 of 396 eligible dates. These daily measurements also change some later positions proposed by a fixed rule that averages recent variance estimates. A common limit on the investment amount can hide much of the difference: for SPY at five minutes, the cap removes 96.95% of the mean gap between uncapped proposals. The expanded comparison shows why that finding should not be generalized to every fund. QQQ still has 50 of 272 evaluation dates with a capped gap of at least five percentage points. EFA shows strong masking at one minute, while incomplete TLT comparisons are withheld. The study provides a reproducible historical example of why data fields, volatility measurements and investment constraints need separate checks. It does not identify true market roughness, establish a universally best cleaning rule or demonstrate trading profits.

Keywords: realized variance; volatility measurement; trade data; rough volatility; investment limits.

Status: computational working-paper draft for human review. Research results are unchanged in this readability revision. AI assistance and data availability are disclosed in Section 8.

Reader's guide

This paper is written for a second-year student who knows basic algebra, averages and variance. Knowledge of high-frequency trading data or stochastic calculus is not needed to follow the main argument. The more advanced derivations remain in Appendix D, where they can be read separately.

The question in everyday language

Imagine that two files describe the same sequence of recorded trades, but store their times and trade labels differently. We use each file to estimate how much prices moved. We then feed each estimate into the same rule for deciding how much to invest. The question is whether the choice of file changes the result, and whether a limit on the investment amount hides some of that change.

The study follows four steps: recorded trades, a daily volatility estimate, an estimate that averages information across days, and a proposed investment amount. It checks each step separately. Agreement in the last step does not automatically mean agreement in the earlier steps.

Terms used throughout the paper

Volatility The size of price fluctuations. Prices can fluctuate strongly even if they finish near where they started.

Return A price change relative to the starting price. A price rising from 100 to 101 has a simple return of 1%. The calculations use log returns, defined in Section 3.

Realized variance (RV) A daily measure made by adding squared price returns. Taking its square root puts it on a volatility scale.

Data source A particular file or representation of the recorded trades. Neither source is treated as a perfectly accurate record of true volatility.

Trade condition A label attached to a trade. A filter uses these labels to decide which recorded trades enter a calculation.

Sampling interval or grid The schedule used to select prices, such as once every minute or once every five minutes.

Estimated state A smoothed variance estimate that combines today's RV with earlier RV values. It is the estimate remembered by the investment rule.

Weight or position The proposed share of capital invested in the fund. A weight of 0.50 means 50% of capital.

Cap The largest position the rule allows. A cap of 1 means at most 100% of capital in the fund.

Origin An evaluation date: the end of a session when information becomes available for the next session's proposed position.

A small example of what a cap can hide

Suppose two estimates lead to proposed investments of 120% and 150% of capital. These are invented amounts, used only to explain the calculation. Before applying a limit, the proposals differ by 30 percentage points. If both proposals face a 100% cap, the final amounts are both 100%. The final difference is zero, even though the estimates behind the proposals differed.

Percent versus percentage points. A change from 50% invested to 55% invested is a difference of 5 percentage points, written 5pp. Relative to the initial 50%, it is a 10% increase. This paper reports differences between investment weights in percentage points. These are not investment returns.

Where to read first

Sections 2–3 explain the data and calculations. Sections 5–6 explain the findings, including the additional funds. Section 7 sets out the limitations. The appendices supply complete tables, detailed mathematics and the record of checks. Codes such as A10 or A13 name stages of the research; they are not models that the reader needs to learn.

Contents

Reader's guide 2

1 Introduction: why the data source matters 5

2 Data and comparison design 5

3 How the measurements and investment rule work 7

4 Simulation: a setting where true volatility is known 11

5 Results for SPY 13

6 Why a cap can hide disagreement 16

7 What the study tells us, and where it stops 20

8 Checks, reproduction and authorship 20

9 Conclusion 21

A Checks and records of the calculations 23

B Complete comparison of the fixed investment rules 24

C Roughness summaries for every chosen lag range 25

D Derivations and interpretation 26

E Reproduction and boundaries 31

F Complete A13 primary counts and slope context 33

1 Introduction: why the data source matters

Volatility measures how strongly prices fluctuate. Researchers and investors often calculate a daily volatility estimate from prices observed during the trading day. That estimate can then enter a forecasting model or a rule for deciding how much to invest. It is therefore worth asking how the estimate depends on the data file used to construct it.

A trade file contains more than prices. It also records when trades occurred and attaches labels describing them. A program may keep some labels and reject others. If two sources store time or labels differently, applying the same filtering instruction can produce different retained observations. The resulting volatility estimates can differ even when the sources contain matching recorded prices, sizes and trading venues.

This paper asks: *when the calculation rule stays fixed, how much does the representation of trade information change a volatility estimate and the investment amount calculated from it?* It also asks whether an upper limit on investment makes two different estimates appear to agree.

We use two historical trade-data representations available through Wharton Research Data Services (WRDS). One records time in seconds; the other records it in milliseconds. We compare the original representations and deliberately exchange their time and condition information. This helps locate which information matters within the chosen calculation. The exercise does not establish which source gives true volatility.

The connection to rough volatility concerns a separate distinction. Prices fluctuate, and the volatility of those fluctuations can itself change. Rough-volatility models describe particularly irregular changes in volatility over short time scales. Gatheral, Jaisson and Rosenbaum [1] provide influential evidence and modeling arguments for this idea. However, true volatility is hidden, so empirical work often studies estimates constructed from prices instead.

Cont and Das [2] show how an observed volatility estimate can look rough even when the underlying process is not rough. Fukasawa, Takabatake and Westphal [3] also study misleading effects of observation error, while developing a different estimation method whose empirical results still support rough volatility. The lesson is that measurement needs attention. These papers do not imply that every low roughness estimate must be an error.

Differences between historical trade databases are already known. Holden and Jacobsen [4] document differences in liquidity measures across Monthly and Daily TAQ. VOLARE [5] discusses source formats, trade timing and trade-size categories when constructing realized measures. This study does not claim to discover that data cleaning matters. It contributes a controlled historical example and follows the effect from the source fields through to a proposed investment.

The main findings are easy to state. Source information changes some daily measurements and the positions proposed from them. A common investment cap can hide much of the resulting difference. The size of the effect also depends on the fund: the relatively small five-minute position gaps for SPY do not carry over uniformly to QQQ. We report the complete fixed comparisons, including small effects and unavailable results.

2 Data and comparison design

2.1 What is observed, and what is a fund-day?

The funds in this study are exchange-traded funds (ETFs): funds whose shares can be traded on an exchange. SPY, QQQ, IWM, EFA and TLT are their trading symbols. Here they identify the

assets whose records we compare; the study does not try to rank the funds as investments.

A *fund-day* means one fund observed during one exchange session. A *source-paired fund-day* means that its records are available in both representations and have passed the required correspondence checks. It is a unit of data coverage, not an independent experiment.

The original source study, labeled A10, covers 398 trading sessions for SPY from 2 January 2013 to 31 July 2014. It also includes QQQ, IWM, EFA and TLT for January 2014. That first export contains 482 fund-days and 121,467,460 matched ordered records before cleaning.

The word *matched* needs care. We check whether the sources have corresponding recorded prices, sizes and venues in the required order. This is weaker than proving that every record represents a different economic trade. Repeated identical records can be difficult to distinguish. Matching also does not prove that either source records every trade perfectly.

A later fixed follow-up, A13, extends all four other funds to the same 398-session calendar. It adds 1,508 paired fund-days and reuses the 84 January pairs already seen. The four-fund extension has 1,592 pairs. Including SPY gives 1,990 distinct fund-days, with reused observations counted once.

The historical window comes from the common period available in the mounted sources. Raw trade records are processed on WRDS rather than downloaded locally. Local files contain daily measures, checks and records of how the files were produced.

2.2 Why some dates are excluded from some results

The mechanism was first examined on two dates: 2 January 2014 and 3 July 2014. These are *discovery dates*: observations that helped motivate the later comparison. Counting them as previously unseen evidence would overstate the strength of that comparison.

The main SPY count of affected dates therefore excludes these two dates, leaving 396 eligible source pairs at one minute. The four-fund follow-up also excludes the January 2014 month, because that panel had already been examined. Its main calendar has 376 eligible dates per fund before any missing measurements. Consequently, the main SPY and follow-up counts have different denominators.

Other calculations ask about a complete history rather than the count on new eligible dates. The descriptive roughness summaries and the smoothed risk rule retain the full calendar, including discovery dates. Removing dates would change the time gaps and the memory of the rule. Each table states which sample it uses.

The design was recorded before each expanded run, but earlier results had already influenced the research question. The follow-up is therefore exploratory historical evidence. It is not untouched chronological validation. Validation observations from 2017–2019 and final observations from 2020 onward remain closed.

2.3 Changing time and trade labels separately

The reference, written M, uses modern time and modern trade-condition labels. The legacy comparison, L, uses legacy time and legacy labels. We also run three mixed constructions. One combines modern time with legacy labels. Another combines legacy time with modern labels. The third keeps legacy time and labels but restores modern category-B information. The rest of the calculation stays fixed. This is a controlled change in the information supplied to the program, rather than a claim about economic causality.

Category B is one of the trade-condition categories. Its documented meaning differs by reporting tape: it denotes average-price trades for CTA issues and bunched trades for UTP issues [6]. A reporting tape is a system for consolidating trade reports. We should therefore not attach one economic meaning to B across all funds.

Restoring B information asks whether that missing information explains a discrepancy in this particular calculation. It does not recommend removing every B-coded trade or adopting the restoration as a general cleaning rule. Direct checks of time rounding and correction fields were also made on the two discovery dates; we do not claim that this stronger field-level check was repeated on every date.

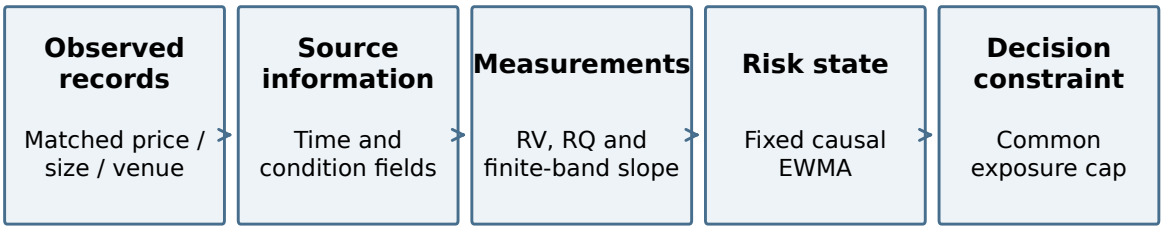


Figure 1: The steps being checked. Recorded trades supply time and condition information. That information enters daily measurements, which enter a smoothed risk estimate and a proposed investment subject to a cap. Matching a later step does not prove that the earlier steps match. This is a calculation chain, not a model of economic cause and effect.

3 How the measurements and investment rule work

3.1 Selecting prices during a session

A trading session is the exchange's open-to-close period. The pipeline first keeps observations with positive finite prices and sizes during that session, applies its rule for uncorrected trades, and retains only condition strings that are blank or contain the regular-trade symbol @ after whitespace normalization.

It then chooses the venue with the largest retained trade count. A venue is an exchange or other trading location. Ties are broken by venue identifier. If several retained prices have the same timestamp, the median price is used. These choices are held fixed across the source constructions.

Next, the program selects prices on six schedules: every 15 seconds, 30 seconds, one minute, five minutes, 15 minutes and 30 minutes. At each scheduled point, it uses the most recent eligible recorded price. This is called *previous-tick sampling*; a tick here is a recorded trade observation.

There is a special rule at the opening point. If no retained trade occurs exactly at the open, the first eligible trade supplies an approximate opening anchor, and the approximation is flagged. Later empty grid points are not filled backwards from a future observation. If a required grid is incomplete, its measurements remain missing. A missing part of the day is not treated as a complete shorter day.

The strict blank-or-@ filter is intentional: it tests what happens when the same literal instruction meets different source information. It retains fewer condition types than the inspected

`highfrequency` cleaner and also differs in the order of venue selection [7]. This is not a full comparison of cleaning packages, nor a full replication of a general BNHLS cleaning procedure.

3.2 Returns, realized variance and quarticity

For prices P_{old} and P_{new}, the log return is

$$r = \log(P_{\text{new}}/P_{\text{old}}).$$

Here log is the natural logarithm. For small price changes, the log return is close to the ordinary percentage change written as a decimal, but they are not exactly equal.

Let t label the session and i label an interval within it. Let M_t be the actual number of sampled return intervals in that complete session. Realized variance (RV) adds squared log returns:

$$\text{RV}_t = \sum_{i=1}^{M_t} r_{t,i}^2, \qquad \text{RQ}_t = \frac{M_t}{3} \sum_{i=1}^{M_t} r_{t,i}^4. \tag{1}$$

The symbol $\sum$ means to add the contribution from each interval. Squaring makes both positive and negative returns contribute positively. Realized quarticity (RQ), the second expression, uses fourth powers and therefore gives relatively more weight to large movements.

Worked example, using invented log returns. Suppose a simplified session has three returns: 0.01, -0.01 and 0.02. Then

$$\text{RV} = 0.01^2 + (-0.01)^2 + 0.02^2 = 0.0006, \qquad \sqrt{\text{RV}} \simeq 0.0245 = 2.45\%.$$

The square root gives a volatility-scale measure of the movements during this invented session. It is not the session's return. Its quarticity is

$$\text{RQ} = \frac{3}{3}\left(0.01^4 + (-0.01)^4 + 0.02^4\right) = 0.00000018.$$

Doubling one return multiplies its squared contribution by four and its fourth-power contribution by sixteen. That explains why RQ can respond strongly to isolated large increments. It does not guarantee that the source difference in RQ is always larger than the source difference in RV.

The factor $M_t/3$ has a mathematical basis. Under an ideal model with independent normal price increments and stated assumptions about the variance path, the fourth moment of a standard normal variable is three. Appendix D.1 derives the normalization carefully. Those assumptions are an explanation of the formula, not a claim that the historical trade data exactly satisfy that model.

The exchange calendar supplies the session length, including early closes, so M_t follows the actual session. Returns are in decimal units and are not annualized at this stage. The measure covers the session and excludes a separately constructed overnight return. Changing decimal returns to percent returns multiplies RV by 100^2 and RQ by 100^4. Units and time coverage must agree before two estimates can be compared.

3.3 When does a source difference count as material?

For a positive valid pair, divide the candidate measurement by the modern reference. A ratio of one means agreement. RV is called *materially different* when the ratio is greater than 1.1 or smaller than 1/1.1. RQ uses the wider limits 2 and 1/2.

For example, invented RV values of 100 and 105 do not pass the RV threshold. Values of 100 and 111 do. A ratio of exactly 1.1 is on the boundary and does not pass the strict threshold. The reciprocal lower limit keeps the decision the same if the two sources exchange places.

For computation, we write the same rule with a log ratio:

$$d_{t,Q} = \log(Q_t^{\text{candidate}}/Q_t^M), \quad |d_{t,\text{RV}}| > \log(1.1), \quad |d_{t,\text{RQ}}| > \log 2.$$

Here Q means either RV or RQ, and vertical bars mean absolute value. These are predefined practical thresholds. Passing one is not the same as statistical significance, and failing one does not establish that two sources are equivalent.

An entry such as 41/396 means 41 material dates out of 396 valid eligible source pairs. The term *incidence* in a table means this fraction of affected dates. A missing pair is excluded from the valid denominator and never counted as agreement.

3.4 A descriptive summary of changes across days

We also ask how daily RV changes over different time gaps. Write $x_t = \log \text{RV}_t$. A lag of one means comparing neighboring trading sessions; a lag of five means comparing sessions five trading days apart.

For each lag k, calculate the average squared change:

$$\widehat{V}(k) = \frac{1}{|\mathcal{I}_k|} \sum_{t \in \mathcal{I}_k} (x_t - x_{t-k})^2, \qquad \widetilde{H}_K = \frac{1}{2} \,\text{slope}_{k=1,\ldots,K}\{\log k, \log \widehat{V}(k)\}. \tag{2}$$

The set $\mathcal{I}_k$ contains dates whose two required RV values are valid; $|\mathcal{I}_k|$ is their count. Thus the first formula says: take the squared change for every valid pair at that lag and average those changes.

The second formula fits a straight line to log average squared change against log lag and takes half its slope. We call the result H-tilde, $\widetilde{H}$. The main calculation uses lags 1–10; fixed alternatives use 1–5 and 1–22. This limited range is what *finite band* means.

A two-lag illustration. If the average squared change doubles when the lag doubles, the log-log slope is one, so H-tilde is one half. If the averages barely increase, the slope is much smaller. This illustration explains the calculation; the reported study uses the stated full lag ranges.

A smaller slope can have several explanations. One is how the underlying volatility changes. Another is measurement error in daily RV. For example, error at both endpoints can add a roughly constant amount to squared changes, making short-lag changes large relative to longer-lag changes. Appendix D.3 derives this under explicit assumptions.

The reported slopes do not identify latent market roughness. Here *latent* means hidden: the true volatility process is not observed. Our statistic summarizes a finite set of daily RV changes. It is not Cont and Das's normalized variation estimator or Fukasawa et al.'s quasi-likelihood estimator, so theorems about those estimators are not transferred to it.

Missing dates retain their place on the calendar. Dropping one and treating its neighbors as consecutive would change the meaning of the lag. When comparing slopes across sources, the paired contrasts also use common valid endpoints so that support differences are visible.

3.5 A risk estimate that remembers previous days

To see whether daily source differences matter later, we use one fixed investment rule. It starts with a smoothed daily variance estimate, s_t. The update is an exponentially weighted moving

average (EWMA):

$$s_0 = \mathrm{RV}_0, \qquad s_t = \lambda s_{t-1} + (1-\lambda)\,\mathrm{RV}_t\,.$$

With the original choice $\lambda = 0.94$, the rule combines 94% of the previous estimate with 6% of today's RV. Repeated updating gradually reduces the influence of older observations.

Worked example, using invented variance values. If the previous estimate is 0.0001 and today's RV is 0.0004, then

$$s_t = 0.94(0.0001) + 0.06(0.0004) = 0.000094 + 0.000024 = 0.000118.$$

A single large source discrepancy can therefore affect many later estimates, even after that day's direct observation has passed. It is important to distinguish one persistent episode from many independent new shocks.

The estimate is initialized at the first day's RV. The first 126 observations are used as a warmup and excluded from the reported evaluation count. This leaves $398 - 126 = 272$ evaluation dates. Complete positive history is required. Missing days are not filled or removed merely to make the update run.

3.6 From estimated variance to a proposed investment

The investment rule aims at a specified volatility target. Let v_* denote that target, and let c denote the position cap. It calculates an uncapped amount u_t and then applies the cap:

$$s_0 = \mathrm{RV}_0, \quad s_t = \lambda s_{t-1} + (1-\lambda)\,\mathrm{RV}_t, \qquad u_t = \frac{v_*}{\sqrt{252 s_t}}, \qquad w_{t+1|t} = \min(c, u_t). \tag{3}$$

The first part repeats the estimate update so that the full rule is shown together. The factor 252 is the fixed annualization convention: daily variance is multiplied by it, then square-rooted to obtain an annual volatility estimate. The target divided by that estimate gives the proposed share of capital. If estimated volatility rises, the proposed position falls.

The function $\min(c, u_t)$ takes the smaller of the cap and the uncapped proposal. Original settings are $\lambda = 0.94$, target $v_* = 0.10$ and cap $c = 1$. The target therefore means 10% annual volatility under this convention, and the cap means at most 100% of capital in the fund.

Continuing the invented variance example gives

$$\sqrt{252(0.000118)} \simeq 0.1724 = 17.24\%, \qquad u_t = \frac{0.10}{0.1724} \simeq 0.580.$$

The rule proposes approximately 58.0% of capital. A 100% cap leaves that proposal unchanged. If two source estimates instead both produced proposals above 100%, the cap would make both final positions equal 100%.

The notation $w_{t+1|t}$ means the proposal for the next session based on information available by the end of session t. An *origin* is that information date, not an independent event. The same session's full RV is never used before the session finishes.

The analysis does not evaluate returns, transaction costs or profit. It compares proposed inputs to a rule. The annualization convention also does not turn open-to-close RV into a complete measure of actual overnight-inclusive portfolio risk. Appendices D.6–D.7 give the exact memory and cap calculations.

A predefined warning criterion is met if at least 2% of valid non-discovery dates are material at a grid, or at least ten evaluated origins have a position difference of at least 5pp. This flags practical sensitivity in the pipeline. It is not a test of originality or proof that one source is more accurate.

3.7 Uncertainty and checks of alternative settings

Neighboring dates may share market conditions or the same persistent source effect. Treating every date as independent would misrepresent how much separate evidence the sample contains. The uncertainty calculation therefore resamples blocks of dates while keeping the source pair together [8].

The stationary bootstrap used here starts at random dates and forms stretches of neighboring dates with random lengths. It repeats the sample construction 3,999 times and recalculates the fraction of material dates each time. Mean block length 22 is the main choice; 11 and 44 are fixed alternatives. All funds use the same resampled date indices, preserving their common calendar dependence.

The displayed intervals summarize variation across these resamples. They rely on the historical window being sufficiently representative and its dependence being compatible with the resampling method. They do not create new market episodes, prove source equivalence or guarantee accuracy after a change in the data-generating regime.

After seeing the original SPY results, the A11 exercise fixes 27 combinations of memory, target and cap:

$$\lambda \in \{.90, .94, .97\}, \qquad v_* \in \{.05, .10, .15\}, \qquad c \in \{.5, 1, 2\}.$$

There are $3 \times 3 \times 3 = 27$ combinations. They are all reported. The exercise is *post-result*: its question was chosen after the original results. It is a sensitivity check and not a new confirmatory sample.

Two useful identities are

$$\frac{u_t^L}{u_t^M} = \sqrt{\frac{s_t^M}{s_t^L}}, \qquad |\min(c, a) - \min(c, b)| \le |a - b|. \tag{4}$$

The first says that the ratio between uncapped proposals depends only on the ratio between their variance estimates: the common target and annualization factor cancel. The second says that applying the same upper cap cannot enlarge a difference. It can shrink that difference, sometimes all the way to zero. Section 6 uses these facts to interpret the findings.

4 Simulation: a setting where true volatility is known

In real trade data we cannot observe the true volatility path directly. A simulation gives a different kind of evidence: we choose a mathematical process, generate artificial observations from it, and know which process produced them. We can then examine what the measurement procedure reports.

The simulation separates three objects. A spot sample is the model's variance at a point in time. Integrated variance adds up variance during a session; its grid approximation uses the average of fine-step variance values on a session clock of length one. RV is then calculated from the artificial price returns. These objects are related, but they are not interchangeable observations.

One model is Ornstein–Uhlenbeck (OU), which fluctuates around a long-run level and has exponentially declining dependence across time. Two further Gaussian models use a generalized Cauchy covariance with local roughness parameters $H = .1$ and $.3$, following the covariance family discussed by Bennedsen, Lunde and Pakkanen [9]. They are not claimed to be fractional OU processes. The OU model's local roughness exponent is .5; this describes its behavior at sufficiently small time scales, not every finite-lag statistic.

For reproducibility, log variance X is converted to positive variance by $v_t = 10^{-4}\exp(X_t - 0.25/2)$. The OU covariance is $0.25\exp(-\kappa|h|)$, where h is a time gap and $\kappa = \log(2)/22$. All settings share the stated 22-day half-correlation scale. This fixes how quickly model dependence declines.

For each model, 128 independent artificial histories contain 504 sessions with 780 fine steps per session. Model variance generates normally distributed price increments conditional on that variance path. Independent log-price noise of 0, 1 or 3 basis points is added at fine-grid price endpoints. A basis point is one hundredth of a percentage point. Coarser returns come from the same underlying fine path, using intervals of 1, 5, 15 and 30 minutes.

The important result is that the same generating process gives different descriptive slopes at different observation stages. Without added price noise, the OU case has mean H-tilde 0.467026 for spot samples, 0.539210 for grid-integrated variance and 0.237906 for five-minute RV. The Monte Carlo standard error of the last mean is 0.002572; it measures simulation sampling uncertainty across the independent artificial histories.

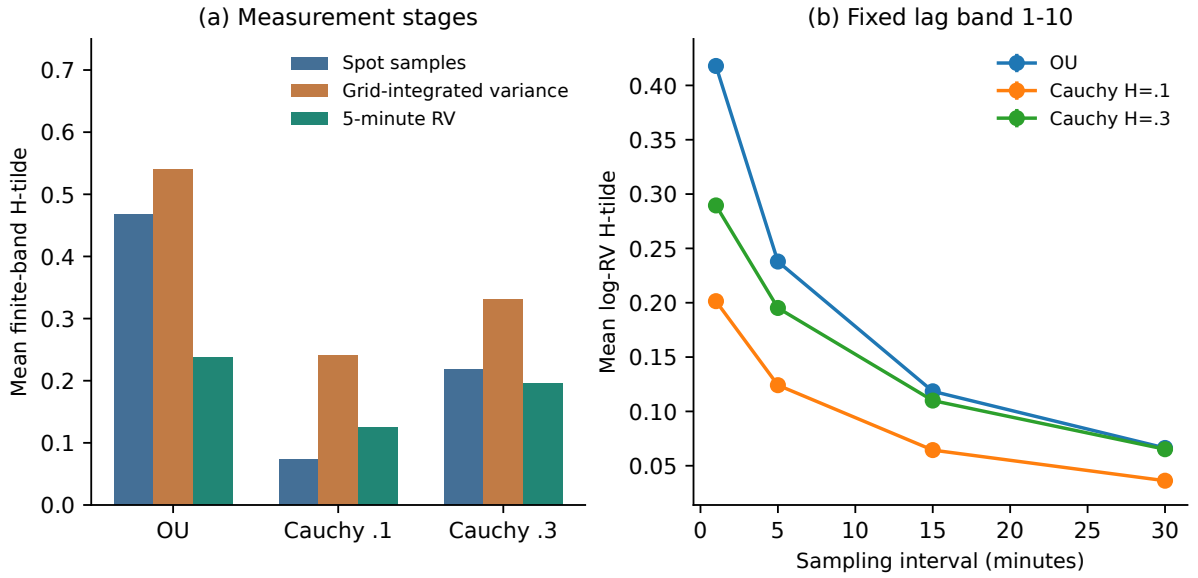


Figure 2: The simulation illustrates the effect of measurement. Panel (a) starts with the same model but compares spot samples, grid-integrated variance and five-minute RV. Panel (b) changes the price-sampling interval. All slopes use daily lags 1–10. Error bars show simulation uncertainty across independent paths, not confidence intervals for real market roughness. No log-price noise is added in this displayed comparison; other fixed settings remain in the supporting tables. Source: A08.

Thus a low slope from observed RV does not by itself reveal the model's local roughness parameter. This is consistent with the measurement concerns in the literature [2, 3]. The simulation is a controlled illustration, not an estimate of the hidden process behind SPY and not a refutation of rough-volatility models.

5 Results for SPY

5.1 Daily measurement differences

At one minute, legacy RV differs materially from modern RV on 41 of 396 valid non-discovery dates. RQ differs materially on 37 of 396. The main block-resampling intervals for the affected fractions are 7.85–12.88% for RV and 7.04–11.65% for RQ.

Table 1: SPY at every fixed sampling interval. RV and RQ entries give affected dates / valid dates, excluding the two discovery dates. H-tilde summarizes the full calendar using lags 1–10. Mean position gaps and counts of origins with a gap of at least 5pp use the original investment rule and 272 evaluation dates.

Grid	RV material	RQ material	$\widetilde{H}_M$	$\widetilde{H}_L$	Mean gap (pp)	$\geq$ 5pp origins
15s	137/396	127/396	0.1703	0.0828	9.225	130
30s	75/396	73/396	0.1629	0.1228	3.261	49
1m	41/396	37/396	0.1542	0.1239	2.632	47
5m	19/396	5/396	0.1409	0.1409	0.032	0
15m	29/396	4/396	0.1046	0.1014	0.010	0
30m	51/396	5/396	0.0936	0.0912	0.014	0

At five minutes, RV still differs materially on 19 dates, while RQ differs on fewer dates. Coarser sampling therefore does not remove every daily discrepancy. The table also shows why one should report every chosen interval: the count does not simply fall as the interval becomes wider.

The one-minute H-tilde values are 0.1542 with modern information and 0.1239 with legacy information. Restoring B information gives 0.1516. These are descriptions of the constructed RV series. They are not estimates that identify the hidden market process. Appendix C includes the other fixed lag ranges so that the interpretation does not depend on one selected band.

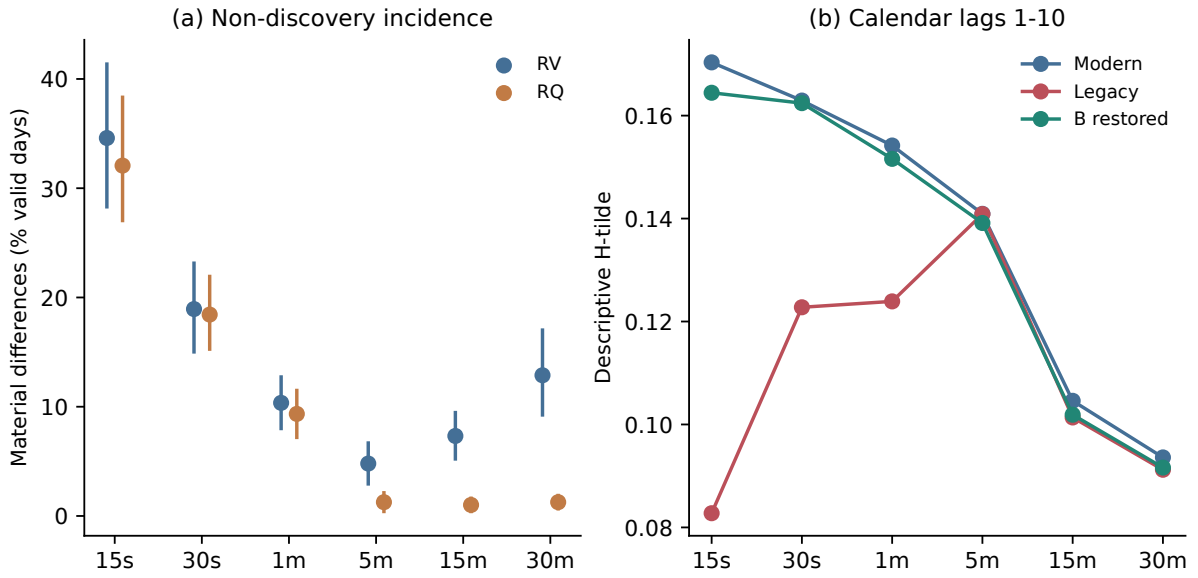


Figure 3: Two different summaries of the SPY comparison. Panel (a) gives the percentage of valid non-discovery dates with material RV or RQ differences and the main block-resampling intervals. Panel (b) gives full-calendar H-tilde, including discovery dates. The panels use different sample rules because they answer different questions. The slopes describe estimated RV rather than true market roughness. Source: A10.

5.2 What changes when time and labels are exchanged?

The crossed constructions help locate the discrepancy. Keeping legacy labels while restoring precise time still leaves 42 material one-minute RV dates. Keeping legacy time but using modern labels reduces that count to 5. Restoring B information under legacy time and the other legacy labels leaves 9.

Table 2: Changing time and condition information separately for SPY at one minute. Every comparison uses the modern reference and 396 valid non-discovery dates. The modern reference compared with itself has zero differences by construction.

Construction	RV material	RQ material	Mean abs. log RV ratio
Legacy time / legacy flags	41	37	0.08524
Modern time / legacy flags	42	35	0.08500
Legacy time / modern flags	5	4	0.02562
Legacy + B information	9	5	0.02744

Within this pipeline, time precision alone therefore explains much less than the available condition information. This is a limited conclusion about the program and source fields. Changing which trades pass a filter can also change the chosen venue and the median price at a timestamp. The effects need not add up as separate economic causes.

The first one-month panel was a small check beyond SPY. After excluding its January discovery date, one-minute RV counts are 4/20 for EFA, 1/20 for IWM, 2/20 for QQQ and 0/20 for TLT. Corresponding RQ counts are 3/20, 1/20, 2/20 and 1/20. These funds share dates, so their observations are not pooled as independent experiments. Section 6.3 reports the much longer fixed follow-up.

5.3 How daily differences affect later proposed positions

Under the original rule, the mean absolute one-minute position gap is 2.632pp and the maximum is 23.9136pp. Absolute means that we measure the size of a gap regardless of which source proposes the larger amount, then average those sizes.

A gap of at least 5pp occurs on 47 of 272 evaluation dates. Those dates form one consecutive episode, from 23 January to 31 March 2014. They are not 47 independent new shocks. The EWMA remembers previous differences, so one period of source disagreement can affect many successive proposals. Restoring B information reduces the mean one-minute gap to 0.0124pp under this particular rule.

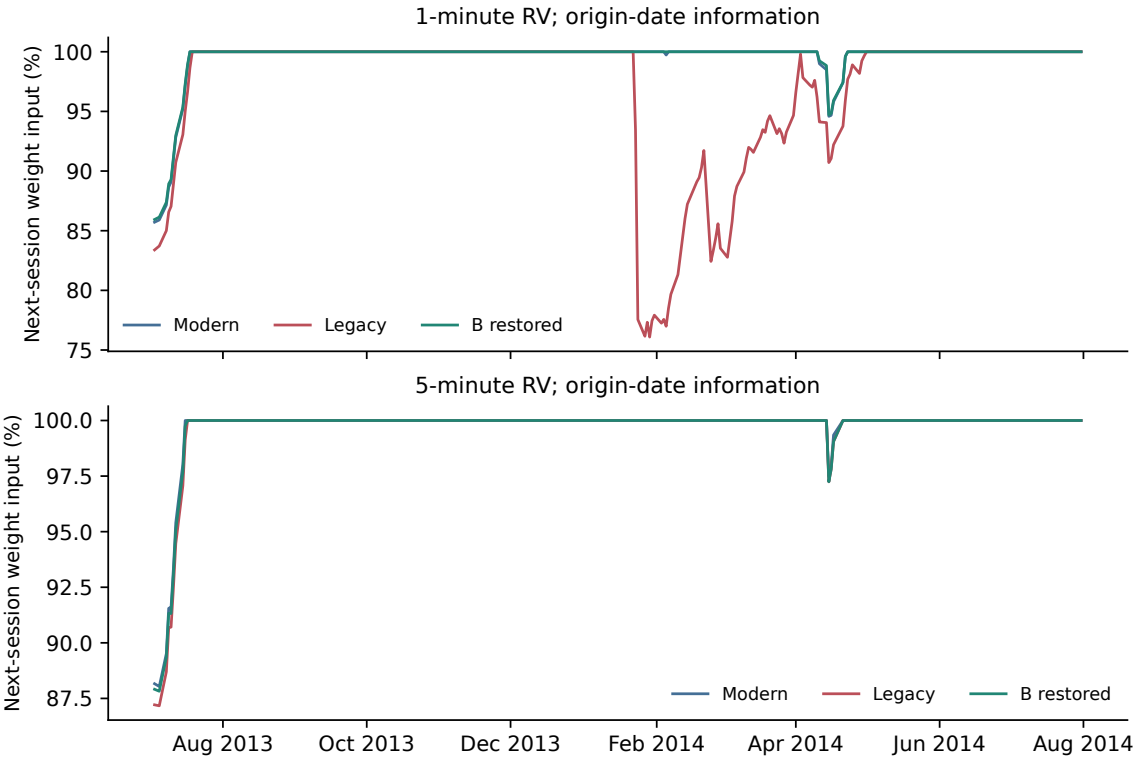


Figure 4: Proposed positions from the original rule: memory $\lambda = .94$, target 10%, cap 100% and a 126-observation warmup. Dates on the horizontal axis identify the completed session supplying information for the next proposal. The one-minute legacy series differs over one persistent episode. The five-minute capped positions look closer, but Section 6 checks whether the cap hides disagreement. These are proposed weights, not realized investment returns. Source: A10.

A further post-result check replaces the legacy RV effect on the two discovery dates with the corresponding modern values. It keeps the dates in place. There are still 46 origins with a gap of at least 5pp. Separately omitting those dates' increment endpoints gives approximate one-minute slopes of 0.1564 modern and 0.1353 legacy. The full result is therefore not entirely due to those two observations. Because the check followed the result, it is a sensitivity exercise rather than independent confirmation or a proposed outlier-cleaning rule.

6 Why a cap can hide disagreement

6.1 A small final gap can come from the common limit

For SPY at five minutes, the mean capped position gap is 0.032pp. Before the cap, it is 1.0555pp. Both source proposals hit the cap on 95.59% of evaluated dates. The cap removes 96.95% of the uncapped mean gap. At one minute it removes 70.38%.

The calculation is

$$\text{fraction of mean gap removed} = 1 - \frac{\text{mean capped absolute gap}}{\text{mean uncapped absolute gap}},$$

when the denominator is positive. It compares mean gap sizes before and after the cap. A large percentage should be read alongside the size of the uncapped gap; it does not measure an investment loss.

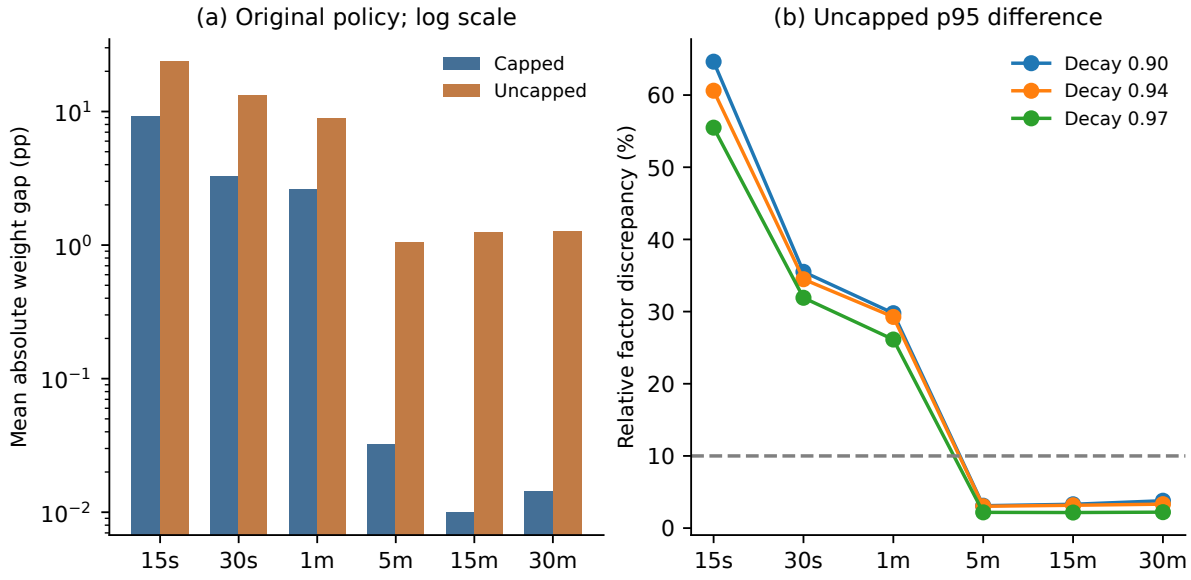


Figure 5: How the cap changes the SPY comparison. Panel (a) shows mean absolute position gaps before and after the cap; its vertical axis is logarithmic, so equal vertical distances mean equal multiplicative changes. Panel (b) summarizes relative uncapped proposal differences at three memory settings. Its p95 is the sample 95th percentile, a description of the observed values rather than an uncertainty interval. Source: A11, chosen after the original result.

This does not mean that the observed difference between one- and five-minute sampling disappears when the cap is removed. At one minute, uncapped proposals differ by a factor above 1.1 on 64–77 of 272 dates across the three memory settings. At five minutes those counts are zero for SPY. The uncapped ratio removes the common target and annualization factor, so it adds useful information about the relative variance estimates. It still does not tell us which estimate is true or what will happen in a future period.

6.2 All fixed rule settings are retained

A policy in this exercise means a specified memory, target and cap. Across all 27 fixed combinations, at least ten origins have a gap of 5pp or more in 18 one-minute policies and 2 five-minute policies. Some policies have zero capped gaps because both proposals reach the same cap.

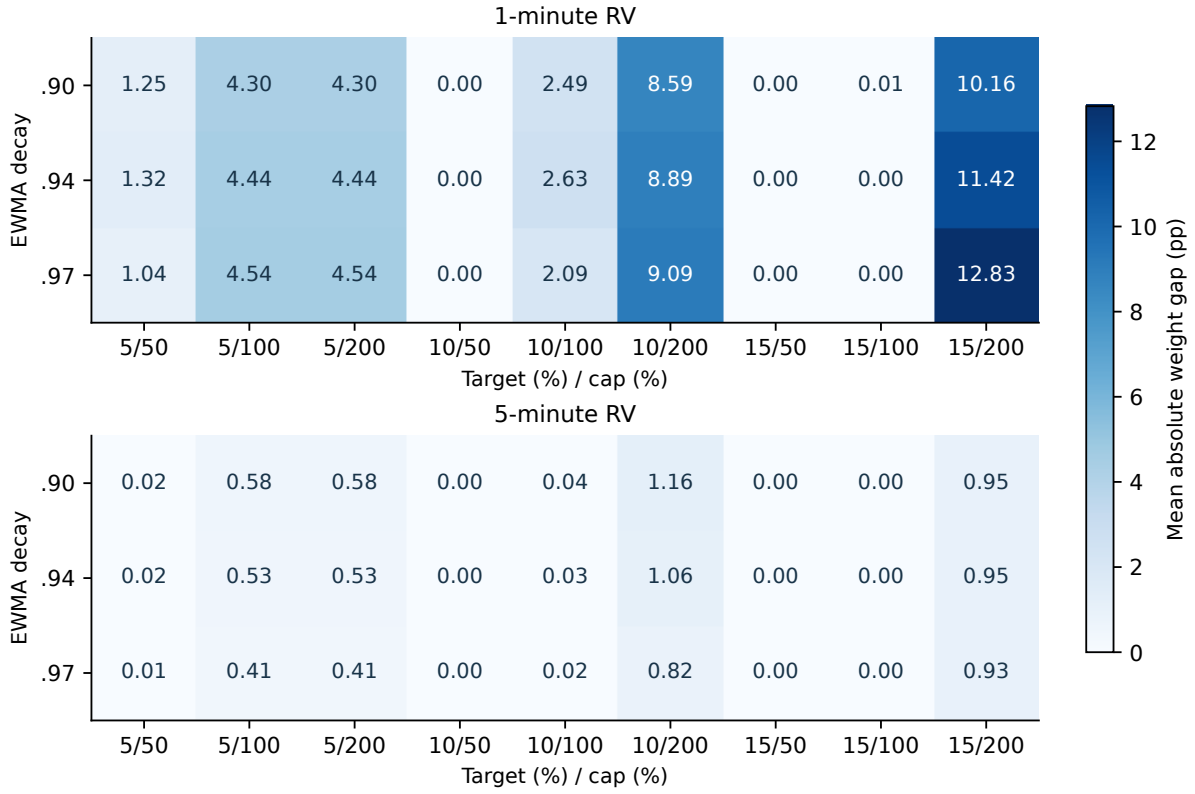


Figure 6: Every fixed combination of memory, target and cap at one and five minutes. Each cell gives the mean absolute position gap in percentage points. Both panels use the same color scale. A colored cell describes one specified scenario; it does not identify a selected or optimal investment strategy. Source: A11.

Appendix B gives the complete table. These policies share data and formulas. They provide sensitivity information, not 27 independent confirmations, and none is selected as best.

An additional calculation asks how concentrated the possible persistence of daily differences is. At memory .94, the largest date accounts for 19.41% of the absolute propagation-potential total, and the five largest account for 66.11%. This calculation adds magnitudes to form an upper-bound measure; signed effects can partly cancel in the actual estimated state. The percentages therefore do not allocate actual investment losses or count independent economic shocks. Appendix D.6 gives the mathematical definition.

6.3 What changes when four more funds are examined?

The follow-up extends QQQ, IWM, EFA and TLT to the same 398-session calendar as SPY. It adds 1,508 paired fund-days and reuses the 84 January observations already examined. That gives 1,592 pairs for the four funds and 1,990 distinct fund-days including SPY.

The follow-up plan was fixed before downloading the additional observations, but after the SPY result was known. This is an exploratory follow-up, not untouched chronological validation. It expands the set of funds within one historical period.

The main count of affected dates excludes the previously seen January 2014 panel and the two discovery dates. That leaves 376 eligible sessions per fund before missing measurements. The full-calendar slopes and smoothed risk calculations retain those dates to preserve time spacing

and memory. The same date indices are used when resampling all funds. Shared dates are not treated as independent evidence merely because several funds are observed.

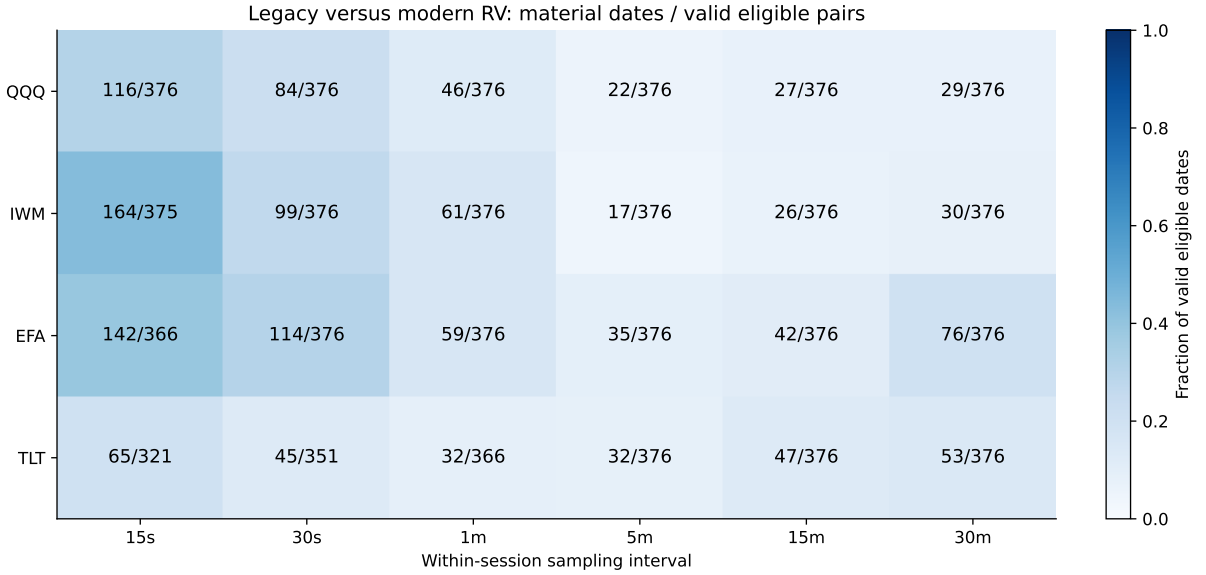


Figure 7: Affected RV dates for every fund and sampling interval in the follow-up. Each cell gives affected dates / valid eligible pairs. Darker cells mean a larger affected fraction; the shared color scale runs from zero to one. The RV ratio must lie outside $[1/1.1, 1.1]$ to count as material. Previously seen January dates and discovery dates are excluded. A missing pair does not count as agreement.

How to read the complete comparison. Table 3 places the daily measurement counts next to the later position differences. These answer different questions. RV/RQ counts concern individual eligible dates. Position gaps use the full chronological history and are evaluated after the warmup. A daily measurement difference can affect later proposals because the rule remembers past RV.

Table 3: All four follow-up funds at one and five minutes under the original rule. RV/RQ entries are affected dates / valid eligible dates. Capped and uncapped columns give mean absolute position gaps in percentage points. The final column counts evaluation dates with a capped gap of at least 5pp. Complete histories have 272 evaluation dates. *Withheld* means that the required TLT history is incomplete; it does not mean a zero effect.

Fund	Grid	RV hits/valid	RQ hits/valid	Capped pp	Uncapped pp	Large origins
QQQ	1m	46/376	34/376	5.8531	7.5218	133
QQQ	5m	22/376	13/376	1.8268	3.4166	50
IWM	1m	61/376	48/376	7.7539	7.7539	186
IWM	5m	17/376	8/376	1.2386	1.2394	9
EFA	1m	59/376	19/376	0.0109	3.2885	0
EFA	5m	35/376	2/376	0.0101	1.0874	0
TLT	1m	32/366	18/366	withheld	withheld	withheld
TLT	5m	32/376	3/376	0.0157	0.8150	0

QQQ qualifies the reassuring SPY result. At five minutes, QQQ has 50 of 272 evaluation dates with a capped gap of at least 5pp; SPY has 0 under the original rule. QQQ's mean capped

gap is 1.8268pp and its uncapped gap is 3.4166pp. The funds use the same calendar, initial-state rule, warmup, target and cap.

Thus the SPY result cannot justify the general claim that five-minute sampling makes this source issue negligible. The comparison concerns proposed amounts from the rule. It does not measure realized losses, and the affected dates are not necessarily independent episodes.

EFA shows the opposite problem: a small final gap hides a larger proposal gap. At one minute, its mean absolute uncapped gap is 3.2885pp. After the cap, it is 0.0109pp. The cap therefore removes 99.67% of the mean gap. This is the same mathematical masking effect as the invented example in the reader's guide.

IWM has a larger one-minute capped gap and much less masking. For TLT, 10 eligible one-minute pairs have incomplete grids, so the specified full-history risk calculation is withheld. Dropping those dates would make nonconsecutive sessions appear consecutive and change the memory update. The study instead keeps the missingness visible.

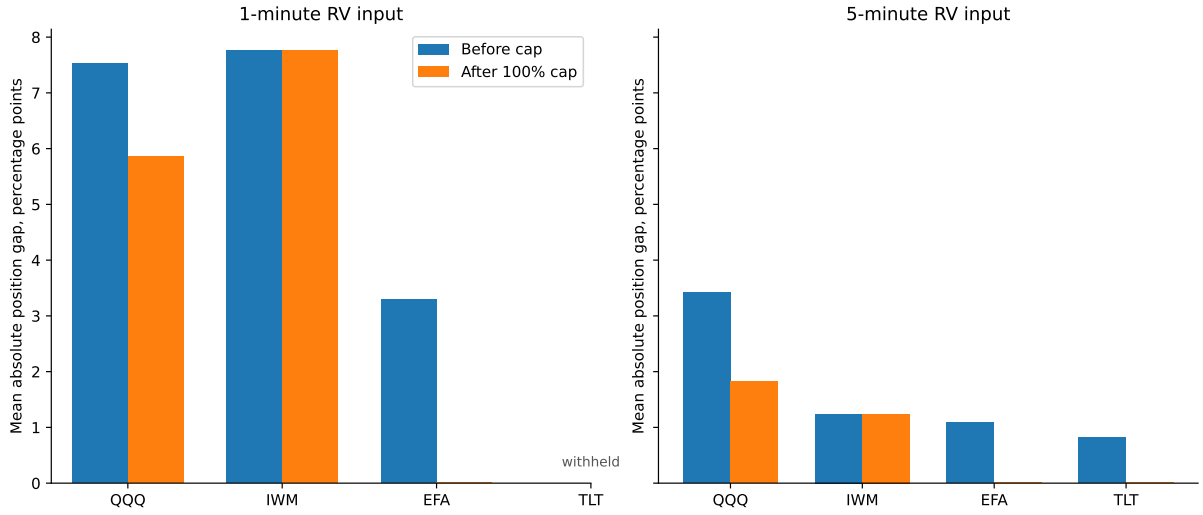


Figure 8: Mean position gaps before and after the common cap. Both panels use the same vertical scale, measured in percentage points. The cap can hide most of the proposal gap for some funds, while leaving much more of it for others. TLT's incomplete one-minute comparison is explicitly withheld. A large percentage of masking must be read together with the actual uncapped gap. These are differences between rule inputs, not trading returns.

The category-B explanation is also fund-specific. For QQQ at one minute, restoring B information leaves 46 material RV dates. Using all modern condition information with legacy time leaves 4. Restoring B therefore does not explain QQQ in the same way as SPY.

This supports checking the full set of source fields and how they interact with the pipeline. It does not establish one economic cause for every fund or one universal category-based repair. The exact meanings of every field in the mounted sources remain a separate documentation question.

Checks and complete reporting. The export passed 6,503 file-member hash checks, calendar and coverage checks, and daily correspondence and measurement checks. A separate result implementation recalculated 6,445 numerical values. It checked the smoothed rule through explicit weighted sums, separately implemented the block resampling and rejected a deliberately changed affected-date count.

Appendix F retains all fixed fund/grid/construction counts and their valid denominators. The

supplied summary files also retain all three slope bands, all risk summaries and all specified block-length intervals. The follow-up shares the same historical source regime as SPY; these checks do not turn it into a test of a new regime or make either source verified economic truth.

7 What the study tells us, and where it stops

The useful distinction is between agreement in the recorded information, agreement in the resulting volatility estimates, and agreement in the final proposed positions. Each needs its own check. A common cap can make the last quantities agree while concealing differences in the estimates that produced them.

The fixed practical warning criterion is met in this historical example. That makes the pipeline sensitive enough to deserve attention under the stated rule. It does not prove a new estimator, academic priority, the best cleaning rule or better investment performance. Earlier research already establishes measurement-induced apparent roughness and historical TAQ compatibility problems.

The historical sample is an important limitation. All five funds share the same source-overlap period. More funds do not provide a new market regime or a test of today's data. Time precision, condition coverage and market structure can change. Block-resampling intervals also rely on assumptions about this historical period; they are not guarantees for a different source regime.

The strict trade filter and approximate opening rule matter too. The filter deliberately asks a narrow question and excludes conditions retained by broader cleaners. Restoring category B is informative for SPY but does not uniformly resolve the other funds. A comparison of alternative cleaning rules would need a separately fixed study.

The modern source is a reference, not verified true integrated variance. Matching recorded rows is not the same as identifying unique economic executions. A lower H-tilde value does not identify hidden roughness, and a smaller investment-weight gap does not prove a better risk estimate.

Finally, the positions are calculated inputs rather than an investment backtest. We do not calculate realized returns, costs, overnight risk or execution outcomes. The original registered 1,250-day forecast comparison remains deferred because the required compatible source history is unresolved. This case study does not replace it with a shorter contest. An option implied-volatility surface is unnecessary for the present question and is not part of the evidence.

8 Checks, reproduction and authorship

8.1 What was checked

Different checks address different possible mistakes. File hashes check whether exported files changed. Coverage checks compare the requested funds and sessions with what was produced. Independent calendar checks detect mistakes such as using a full-session interval count on an early-close day.

The daily RV/RQ calculations are compared between SQL aggregation and a separate NumPy construction. Recorded-row correspondence is checked separately. Two programs agreeing on RV would not, on its own, prove that they used the intended trading session or matching source records.

At the result level, a separate implementation recalculates 900 original A10 values. The A11

check recalculates 10,584 values and 96 values that overlap A10. It uses a direct weighted sum to check the EWMA update instead of repeating the analysis's recursive calculation. It also rejects a deliberately altered policy output. The A13 checks are described in Section 6.3.

The mathematical appendix has separate checks of selected formulas, including normal moments, variance scaling, memory initialization and cap boundaries. These checks establish agreement for the stated cases and assumptions. They do not prove that every possible error is absent or that the assumptions hold for market data. Appendix A records the checks and their limits.

8.2 How numbers in the paper are connected to outputs

The displayed research numbers and table cells are generated from recorded analysis outputs. A separate checker follows each number back to its specified source field and checks its transformation and printed value. A *run ID* is a unique label for a recorded execution; Appendix A and the repository use these labels to identify the relevant outputs.

This makes the calculations auditable without asking the reader to trust a manually typed table. The source code, file hashes and recorded configurations support reproduction. The data-free synthetic examples can be run without a WRDS account. Recreating the historical empirical results still requires the separately licensed input observations.

Data availability. The repository supplies original code, synthetic examples, aggregate research summaries and the manuscript. Raw trade records, licensed daily observations and private working Git history are excluded. Closed validation and final-period observations have not been used.

AI and authorship disclosure. Martin Valachy set the research direction, scope and goals. Codex implemented the computational pipeline, numerical checks, figures and draft. Human comprehension and independent human peer review are not asserted complete. Agreement between separate programs is computational checking, not human peer review. The paper is a working draft for honest undergraduate research review; it does not imply journal acceptance or work performed without AI assistance.

9 Conclusion

In this historical comparison, the way trade information is stored changes some volatility estimates and the investment positions proposed from them. Applying the same filtering code does not ensure that different data sources preserve the same information.

An upper position limit adds a second issue: similar final positions can hide disagreement between the underlying estimates. The additional funds make this conclusion more specific. QQQ retains substantial five-minute position differences, while EFA shows strong one-minute cap masking. Incomplete TLT histories are left unavailable rather than turned into apparent agreement.

The practical lesson is to check source information, daily measurements, smoothed estimates and capped positions separately. The study supports that audit procedure in the stated historical setting. It does not identify true market roughness, select a universally best sampling interval or demonstrate a profitable strategy.

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A Checks and records of the calculations

Table 4: What was checked, and what remains outside that check.

Component	Evidence	Limit
Source/sample	Fixed calendar/panel, exported member hashes, per-file source alignment flags	Identity is observable-row correspondence, not unique execution identity
RV/RQ construction	SQL versus NumPy; exact grid counts; missingness; synthetic unit/early-close errors	Does not certify a universally best trade filter or true integrated variance
Finite-band slopes	Independent log-variogram/OLS arithmetic; scale invariance; calendar holes	No local market-H identification or empirical H coverage claim
EWMA and weights	Recurrence versus direct convolution; causality, initialization and fixed-policy overlaps	No return, cash, overnight, cost or execution evaluation
Policy/influence	Every fixed setting; explicit shock sums; altered summary rejected	Post-result; no policy-selection or nonlinear causal-loss attribution
Synthetic baseline	Known covariance/measurement checks; independent table arithmetic; coupled refinement	Conditional simulator evidence, not empirical model validation
Paper/presentation	Source-linked numeric facts, compilation, references and visual inspection	Cannot substitute for human comprehension and research review

The primary provenance anchors are:

A08 simulation `side_project_study_20261021T214152Z`.

A08 independent tables `side_project_table_check_20261021T215134Z`.

A08 refinement `side_project_refinement_20261021T214206Z`.

A10 full analysis `source_replication_analysis_20261022T014736Z`.

A10 independent results `source_result_check_20261022T015029Z`.

A10 export provenance `source_export_verified_20261022T014656Z`.

A11 sensitivity `source_policy_robustness_20261022T032458Z`.

A11 independent results `source_policy_check_20261022T032503Z`.

Derivation checks `paper_derivation_check_20261022T121236Z`.

The source configurations and methods are recorded in `configs/side_project.json`, `configs/source_replication.json`, `configs/source_policy_robustness.json` and the associated A08/A10/A11 protocol files. A11 was frozen locally before computing its sensitivities, after the original A10 outcome was known. The exploratory chronology is retained rather than relabeled as a confirmatory design.

B Complete comparison of the fixed investment rules

Table 5: Complete A11 policy matrix. Targets/caps are percentages; gaps are mean absolute weight differences in pp; "large" counts origins with an absolute gap of at least 5pp. All rows use the same 272 dependent origins.

λ	Target %	Cap %	1m gap (pp)	1m large	5m gap (pp)	5m large
0.90	5	50	1.247	26	0.021	0
0.90	5	100	4.296	92	0.578	1
0.90	5	200	4.296	92	0.578	1
0.90	10	50	0.000	0	0.000	0
0.90	10	100	2.494	37	0.042	0
0.90	10	200	8.592	130	1.157	9
0.90	15	50	0.000	0	0.000	0
0.90	15	100	0.010	0	0.000	0
0.90	15	200	10.162	116	0.954	11
0.94	5	50	1.316	29	0.016	0
0.94	5	100	4.443	86	0.528	0
0.94	5	200	4.443	86	0.528	0
0.94	10	50	0.000	0	0.000	0
0.94	10	100	2.632	47	0.032	0
0.94	10	200	8.885	147	1.055	7
0.94	15	50	0.000	0	0.000	0
0.94	15	100	0.000	0	0.000	0
0.94	15	200	11.422	142	0.949	13
0.97	5	50	1.044	19	0.011	0
0.97	5	100	4.543	77	0.409	0
0.97	5	200	4.543	77	0.409	0
0.97	10	50	0.000	0	0.000	0
0.97	10	100	2.088	55	0.023	0
0.97	10	200	9.087	183	0.818	0
0.97	15	50	0.000	0	0.000	0
0.97	15	100	0.000	0	0.000	0
0.97	15	200	12.826	173	0.931	1

C Roughness summaries for every chosen lag range

Table 6: Full-window finite-band H-tilde at every grid and fixed band. Both explicit discovery dates remain included. These values are descriptive and do not carry latent-H confidence intervals.

Largest lag	Grid	Modern	Legacy	B restored
5	15s	0.1921	0.0913	0.1800
5	30s	0.1866	0.1532	0.1839
5	1m	0.1773	0.1546	0.1763
5	5m	0.1742	0.1708	0.1709
5	15m	0.1538	0.1492	0.1518
5	30m	0.1389	0.1370	0.1386
10	15s	0.1703	0.0828	0.1644
10	30s	0.1629	0.1228	0.1624
10	1m	0.1542	0.1239	0.1516
10	5m	0.1409	0.1409	0.1391
10	15m	0.1046	0.1014	0.1019
10	30m	0.0936	0.0912	0.0916
22	15s	0.1438	0.0766	0.1397
22	30s	0.1342	0.1027	0.1331
22	1m	0.1253	0.1046	0.1228
22	5m	0.1021	0.1039	0.1014
22	15m	0.0802	0.0772	0.0783
22	30m	0.0643	0.0618	0.0627

D Derivations and interpretation

This appendix supplies the mathematical bridge from a sampled price to a reported decision input. The arguments below are conditional calculations under stated assumptions, not fitted models of the historical sample. They explain what the diagnostics can reveal and why the empirical comparisons do not identify latent market volatility.

Reading the mathematics. This optional appendix uses probability and calculus beyond the algebra needed for the main text. $\mathbb{E}$ means an average over repeated outcomes in the model; Var measures variation around that average. A statement conditional on a variance path holds that path fixed while averaging over price shocks. The symbol $\sum$ adds discrete terms, while an integral adds contributions continuously over time. The continuous-time model motivates the calculation; the finite-grid steps show exactly how the displayed RV and RQ formulas arise.

D.1 From price increments to realized variance and quarticity

Purpose. We ask what squared and fourth-power returns average to under an ideal model. This explains the measurement formulas without assuming that a real trade file perfectly satisfies the model. The central step is to write an interval return as its standard deviation times a standardized random shock.

Use a trading-session clock $u \in [0, 1]$, with the actual session rescaled to unit length. In an ideal continuous-price model,

$$dp_u = b_u\, du + \sigma_u\, dW_u, \qquad IV = \int_0^1 \sigma_u^2\, du, \qquad IQ = \int_0^1 \sigma_u^4\, du. \tag{5}$$

Here p is log price, IV is integrated variance and IQ is integrated quarticity. Quadratic-variation theory motivates sums of squared returns; the continuous stochastic-volatility setting supplies additional distributional results. Barndorff-Nielsen and Shephard [10] distinguish this consistency statement from assumptions needed to quantify precision (their Sections 1 and 3). The following finite-grid calculation makes the moment normalizations explicit.

For this calculation, ignore drift and suppose the variance path is deterministic, or independent of the Brownian price shocks. On a grid of M intervals, write

$$r_i = \sqrt{a_i} Z_i, \quad a_i = \int_{(i-1)/M}^{i/M} \sigma_u^2\, du, \quad Z_i \overset{\text{iid}}{\sim} N(0, 1). \tag{6}$$

The assumptions imply independent conditional increments. They are stronger than a general model with leverage; conditional Gaussianity is not asserted for arbitrary stochastic volatility. Since $\mathbb{E}Z_i^2 = 1$ and $\mathbb{E}Z_i^4 = 3$,

$$\mathbb{E}[\text{RV} \mid a] = \sum_i a_i = IV, \tag{7}$$

$$\text{Var}(\text{RV} \mid a) = 2 \sum_i a_i^2, \tag{8}$$

$$\mathbb{E}[\text{RQ} \mid a] = M \sum_i a_i^2, \qquad \text{RQ} = \frac{M}{3} \sum_i r_i^4. \tag{9}$$

The variance uses $\text{Var}(Z_i^2) = 3 - 1 = 2$ and vanishing cross covariances. For a sufficiently regular variance path, the last expectation is a Riemann approximation to IQ. With constant variance $a_i = IV/M$, it is exactly IV^2. This derives the factor $M/3$ rather than treating it as a

vendor convention. Under the stated finite-grid model, $2\,\mathrm{RQ}/M$ is an unbiased estimator of the conditional variance in Equation (8).

Quarticity responds particularly strongly to isolated increments: multiplying one return by k multiplies its RV contribution by k^2 and its RQ contribution by k^4. That does not mean that a larger source RQ discrepancy always occurs: retained trades, signs, neighboring returns and grid endpoints all matter. The complete incidence table is therefore necessary.

The session clock also explains why M must follow the actual session. If a complete shortened session has m returns but its quarticity is incorrectly multiplied by a full-session count M, the computed RQ is M/m times the intended session-normalized RQ, even when the prices and returns are identical. Missing intervals cannot be repaired by substituting the observed count into the formula: the resulting returns may cover an irregular or truncated interval. This is why incomplete grids are withheld rather than silently renormalized.

For decimal log returns, RV has squared-return units and RQ has fourth-power units. Converting all returns to percentage units multiplies RV by 100^2 and RQ by 100^4; changing the currency denomination of every price leaves log returns unchanged. Multiplying a daily RV by an annualization factor A is another explicit transformation. A daily RQ can be multiplied by A^2 to express the corresponding fourth-power scale; it does not then become an estimate of the integral of quarticity over an observed year. Units and observation windows must both agree before reconciliation.

D.2 What a relative measurement diagnostic does and does not estimate

Purpose. We compare uncertainty in RV with the size of RV itself. This produces a scale-free ratio. Because its denominator is also random, the average of the ratio need not equal the ratio of two averages. That distinction explains the finite-sample result below.

The preceding conditional-variance identity motivates the dimensionless diagnostic

$$\widehat{q} = \frac{2\,\mathrm{RQ}}{M\,\mathrm{RV}^2} = \frac{2}{3}\frac{\sum_i r_i^4}{(\sum_i r_i^2)^2}. \tag{10}$$

A first-order delta approximation, $\log(\mathrm{RV}) - \log(IV) \simeq (\mathrm{RV} - IV)/IV$, connects a relative variance to log measurement error when that error is small. It does not establish iid additive errors across days, eliminate leverage, handle market microstructure noise or certify that $\widehat{q}$ itself is an unbiased log-error variance.

Even the constant-variance Gaussian case illustrates the distinction between a ratio of expectations and an expectation of a ratio. Let $P_i = Z_i^2/\sum_j Z_j^2$. Spherical symmetry, or the Dirichlet law with all parameters $1/2$, gives

$$\mathbb{E}[P_i^2] = \frac{(1/2)(3/2)}{(M/2)(M/2+1)} = \frac{3}{M(M+2)}, \qquad \mathbb{E}[\widehat{q}] = \frac{2}{M+2}. \tag{11}$$

In contrast, $\mathrm{Var}(\mathrm{RV} \mid IV)/IV^2 = 2/M$. The finite-sample discrepancy is a consequence of the random denominator. This diagnostic is supporting methodology, not an extra empirical endpoint of A10/A11. In particular, no empirical variogram is "corrected" by subtracting it.

D.3 Two different places where observation noise enters

Purpose. Error can enter the recorded price before returns are calculated, or enter a daily volatility estimate after prices have been aggregated. These are different assumptions. We derive each effect separately so that a formula for one is not silently applied to the other.

First consider price-level noise on a fixed grid: $y_i = p_i + \eta_i$, where independent centered errors have variance ω^2 and are independent of the efficient price. Observed returns are $\widetilde{r}_i = r_i + \eta_i - \eta_{i-1}$. Expansion gives

$$\mathbb{E}[\widetilde{\mathrm{RV}} \mid a] = IV + 2M\omega^2. \tag{12}$$

The cross term has zero expectation and each differenced error has variance $2\omega^2$. Adjacent error differences are negatively correlated; independence across observed returns is unnecessary for this expectation. If the level noise is also Gaussian, each marginal observed return is Gaussian with variance $a_i + 2\omega^2$, hence

$$\mathbb{E}[\widetilde{\mathrm{RQ}} \mid a] = M\sum_i (a_i + 2\omega^2)^2 = M\sum_i a_i^2 + 4M\omega^2 IV + 4M^2\omega^4. \tag{13}$$

Higher sampling frequency can therefore increase contamination even while it improves the ideal noise-free discretization. This illustrative model is implemented in the supporting simulation; it is not a claim that the historical source discrepancy consists of independent Gaussian quote errors. In particular, source-specific condition information is structured, not an iid perturbation.

The second model concerns a daily log-variance proxy: $Y_t = X_t + e_t$. If e_t is centered, iid, independent of X, and has variance q, distinct endpoints give

$$V_Y(k) := \mathbb{E}[(Y_{t+k} - Y_t)^2] = V_X(k) + 2q. \tag{14}$$

To derive this, expand the square, drop the cross term by independence, and use $\mathrm{Var}(e_{t+k} - e_t) = 2q$. For stationary but correlated independent measurement error, the addition instead equals $2[\gamma_e(0) - \gamma_e(k)]$. If the error is related to X, an additional cross term remains. Price-level noise and daily log-proxy noise therefore cannot be identified with each other merely because both can flatten a plot.

D.4 Why the finite-band statistic is not latent roughness

Purpose. A local roughness exponent describes behavior as a time gap becomes very small. Our fitted slope instead summarizes a fixed range of daily gaps. The equations below show how noise and the curvature of a model's variance-of-changes function can separate those two quantities.

For a positive population variogram, define a local scale diagnostic

$$H_{\mathrm{eff}}(\tau) = \frac{1}{2}\frac{d\log V(\tau)}{d\log \tau} = \frac{\tau V'(\tau)}{2V(\tau)}. \tag{15}$$

If $V(\tau) = C\tau^{2H}$ exactly, this equals H at every scale. Adding the iid daily log-noise floor produces

$$H_{\mathrm{eff},Y}(\tau) = H\frac{C\tau^{2H}}{C\tau^{2H} + 2q}. \tag{16}$$

For positive q, the factor lies between zero and one. This is a precise attenuation mechanism under this model, not a proof that every empirical low slope is noise.

For a stationary OU log-variance process with covariance $v\exp(-\kappa|\tau|)$, stationarity gives $V_X(\tau) = 2v[1 - \exp(-\kappa\tau)]$. Differentiating yields

$$H_{\mathrm{eff},X}(\tau) = \frac{\kappa\tau}{2(\exp(\kappa\tau) - 1)}, \qquad H_{\mathrm{eff},Y}(\tau) = \frac{\tau v\kappa\exp(-\kappa\tau)}{2v[1 - \exp(-\kappa\tau)] + 2q}. \tag{17}$$

Since $\exp(z) - 1 \sim z$ near zero, the noise-free local limit is $1/2$. It falls at longer scales because OU increments approach their stationary variance. An OU process is locally diffusive, not

differentiable. "Nonrough" here distinguishes it from a volatility model with a smaller local exponent; it does not mean a smooth deterministic path. A finite nonzero noise floor further changes the small-scale behavior.

The reported estimator uses finitely many calendar lags, not a derivative at zero. With $x_k = \log k$, $z_k = \log \widehat{V}(k)$, its explicit formula is

$$\widetilde{H}_K = \frac{1}{2} \frac{\sum_{k=1}^{K} (x_k - \bar{x})(z_k - \bar{z})}{\sum_{k=1}^{K} (x_k - \bar{x})^2}. \tag{18}$$

It recovers H for an exact power law on that band, but otherwise summarizes the curvature and errors at the selected lags. Lagged increments overlap, their estimates are dependent, and different lags can have different valid endpoint counts. Ordinary regression standard errors would not automatically be valid. Missing calendar dates are not compressed, because doing so changes the meaning of k. The known-truth experiment isolates this distinction without fitting a market model.

Multiplying every daily RV by the same positive constant adds a constant to every log RV and leaves all increments and $\widetilde{H}_K$ unchanged. A source discrepancy that changes the slope must therefore have more structure than a constant variance-unit multiplier. This invariance does not identify which source is economically preferable.

D.5 Controlled source contrasts and interactions

Purpose. Changing a time field can have a different effect depending on which condition field accompanies it. The interaction term records this dependence. It explains why counts from separate changes cannot simply be added to assign economic causes.

Let $F(T, C)$ be any positive daily measurement produced by the entire fixed pipeline from time representation T and condition representation C. The crossed construction changes inputs to this function while retaining the observed price/size/venue sequence. On log scale, define

$$D_T = \log F(L, M) - \log F(M, M), \tag{19}$$

$$D_C = \log F(M, L) - \log F(M, M), \tag{20}$$

$$I = \log F(L, L) - \log F(L, M) - \log F(M, L) + \log F(M, M). \tag{21}$$

Then the legacy/reference log difference equals $D_T + D_C + I$ by algebra. An additive representation would set $I = 0$, but the pipeline provides no reason to assume this. Changing conditions can change retained-venue counts, the selected venue, same-time medians and previous-tick endpoints. Moreover, the materiality indicator thresholds an absolute difference, so incidence counts cannot be added as if they were signed effects. The paper reports the complete crossed constructions rather than claiming an additive causal decomposition. B restoration is a further controlled diagnostic within this function, not a universally justified trade deletion.

D.6 State propagation, initialization and persistence

Purpose. Repeated substitution turns the update rule into an explicit weighted sum of the initial observation and later observations. This shows why an old difference can remain in today's estimate, and why the first observation needs a different coefficient from later updates.

Starting from $s_0 = \mathrm{RV}_0$, repeated substitution into Equation (3) gives

$$s_t = \lambda^t \,\mathrm{RV}_0 + (1 - \lambda) \sum_{i=1}^{t} \lambda^{t-i} \,\mathrm{RV}_i \,. \tag{22}$$

The coefficients sum to one. The initial coefficient is λ^t, not $(1-\lambda)\lambda^t$; treating initialization like a later shock would lose mass. This direct convolution is the independent check on the recursive implementation.

For source differences $\delta_i = \mathrm{RV}_i^L - \mathrm{RV}_i^M$, subtraction gives exactly the same linear filter of δ_i. If only a later observation $j > 0$ differs, its effect at $t \geq j$ is $(1-\lambda)\lambda^{t-j}\delta_j$. Its half-life, in sessions, is $\log(1/2)/\log\lambda$. One discrepant input can therefore create many successive differing states; counting affected origins does not count independent source shocks. The paper labels weights by their information origin: s_t only supplies the session-$t+1$ input, avoiding use of the same day's complete RV before it exists.

For evaluated origins $\mathcal{T}$, put $a_{t,0} = \lambda^t$ and $a_{t,i} = (1-\lambda)\lambda^{t-i}$ for $1 \leq i \leq t$. The triangle inequality yields

$$\sum_{t\in\mathcal{T}} |s_t^L - s_t^M| \leq \sum_i |\delta_i| \sum_{\substack{t\in\mathcal{T} \\ t\geq i}} a_{t,i}. \tag{23}$$

The right-hand summands define the absolute propagation-potential shares. They omit signed cancellation and precede the nonlinear state-to-weight transformation. Their concentration is informative about sensitivity, but is not an allocation of actual portfolio losses.

D.7 From variance states to weights and cap masking

Purpose. The rule first takes a square root to convert variance to volatility, then divides the target by that volatility to obtain an investment amount. We examine how differences pass through those operations and when a shared cap makes two final amounts equal.

Let $s_t > 0$ be a daily variance state, A a common annualization factor, $v_* > 0$ an annual volatility target and $c > 0$ a cap. Equating an unconstrained position's estimated annual volatility to its target gives

$$u_t\sqrt{As_t} = v_*, \qquad u_t = \frac{v_*}{\sqrt{As_t}}, \qquad \frac{\partial \log u_t}{\partial \log s_t} = -\frac{1}{2}. \tag{24}$$

The sensitivity follows by differentiating $\log u_t = \log v_* - (\log A + \log s_t)/2$. A small relative variance change therefore produces approximately half as large an oppositely signed relative unconstrained weight change. The exact ratio is

$$\frac{u_t^L}{u_t^M} = \sqrt{\frac{s_t^M}{s_t^L}}, \qquad \left|\log\frac{u_t^L}{u_t^M}\right| = \frac{1}{2}\left|\log\frac{s_t^L}{s_t^M}\right|. \tag{25}$$

Common targets and annualization cancel. If an entire source RV path is multiplied by $a > 0$, consistent initialization makes its state exactly a times as large and its unconstrained weight $a^{-1/2}$ times as large. Thus a constant unit error leaves the roughness slope unchanged while changing the risk input, illustrating why both audits are needed.

Now set $w_t = \min(c, u_t)$. The states at which a weight saturates satisfy

$$s_t \leq \frac{v_*^2}{Ac^2}. \tag{26}$$

Within this region the constrained weight is locally insensitive to the state. If both sources are there, their weights agree exactly regardless of their state ratio. If neither is there, capping changes neither weight. If just one is there, the cap reduces the gap. These three exhaustive cases prove

$$|\min(c,a) - \min(c,b)| \leq |a-b| \qquad (a,b>0). \tag{27}$$

The aggregate attenuation is defined as one minus the mean capped absolute gap divided by the mean uncapped absolute gap, when the latter is positive. Its value depends on the chosen target and cap. This makes the uncapped state-ratio diagnostic necessary for interpreting apparent decision agreement. It does not turn either estimated state into true risk or establish an optimal portfolio policy.

D.8 What the block intervals quantify

Purpose. We explain how the affected-date fraction is recalculated after blocks of calendar dates are sampled. Keeping the numerator, valid denominator and source pairs on the same sampled dates preserves the intended comparison. It does not create fresh independent market episodes.

For a source pair and grid, let J_t indicate a valid comparison and I_t indicate both validity and a discrepancy above the fixed threshold. The empirical incidence is $\widehat{p} = \sum_t I_t / \sum_t J_t$, with the two discovery dates ineligible for both sums. Resampling the numerator and denominator on the same calendar indices preserves this definition; resampling only discrepancies or compressing to valid dates changes the experiment.

The stationary bootstrap [8] starts at a uniformly drawn calendar index. Each subsequent index restarts uniformly with probability $1/L$, or continues to the next circular calendar index otherwise. Thus $\Pr(B = b) = (1 - 1/L)^{b-1}/L$ for block length B, with mean L. Using identical indices across paired objects retains their contemporaneous dependence. Recomputing the incidence on each resample yields the displayed percentile intervals; block-length sensitivities assess one aspect of dependence handling.

These intervals rely on a useful stationary, weak-dependence approximation and the representativeness of the available window. They do not supply new independent episodes, remove a source-regime change, establish source equivalence or account for every exploratory choice. The slope panels are descriptive and are not assigned these incidence intervals. The policy matrix is a systematic post-result sensitivity, not a family of successful independent hypothesis tests.

E Reproduction and boundaries

The private original A10 export is kept outside the repository. From the authorized research workspace, the empirical pipeline is checked using `verify_source_export.py`, `analyze_source_replication.py`, `check_source_results.py`, `analyze_policy_robustness.py` and `check_policy_robustness.py`. Each receives the private-folder argument. The paper preparation script reads these checked outputs, writes named facts/tables and verifies the displayed weight paths against the original summary. A separate checker reads the source records using standard-library arithmetic and compares the generated fact macros.

The PDF is compiled from editable LaTeX sources. The delivery includes a claim map, bibliography and verification summary; the wider local atlas is supplementary and not a substitute for the paper's argument. The portable synthetic demo has its own isolated installation procedure. Reproduction generates new run IDs and timing metadata; scientific tolerances, not byte equality across numerical platforms, determine arithmetic agreement. No older mixed-period audit script is part of these instructions.

A13 is reproduced from its separate licensed aggregate export using `verify_etf_breadth.py`, `analyze_etf_breadth.py` and `check_etf_breadth.py`. The exported-member fingerprint, fixed calendar and previously seen January hashes are checked before analysis. `prepare_breadth_`

`paper.py` reads only its checked summary tables; `check_breadth_paper.py` independently checks every additional numerical macro and catches a planted altered fact. The complete summary tables accompany the manuscript without the underlying daily observations.

The original broader forecast contest and unresolved source-reconciliation gates are preserved as deferred historical plans. No shortened-window forecast winner is reported. The option-surface extension was parked after clarification of the paper's scope, and no option observations were acquired. No final-period, publication, spending or contact action is required by this reproduction path.

F Complete A13 primary counts and slope context

This appendix retains every fund, grid and construction. "Time only" means modern time with legacy flags; "Cond. only" means legacy time with modern flags. B restoration is a diagnostic, not a production rule. Every incidence entry is material/valid eligible pairs. The primary eligible calendar has 376 dates per fund; valid denominators can be smaller. January 2014 is previously observed and is excluded here, together with the fixed discovery dates. Missingness is not classified as agreement.

The frozen design is `protocol/A13_ETF_BREADTH.md`, with configuration `configs/etf_breadth.json`. Its provenance anchors are:
Export check `etf_breadth_export_verified_20261022T144522Z`.
Analysis `etf_breadth_analysis_20261022T144854Z`.
Independent results `etf_breadth_independent_check_20261022T145028Z`.

Table 7: Complete A13 RV primary incidence. Materiality is an absolute log ratio above log(1.1), equivalently a ratio outside $[1/1.1, 1.1]$. Modern/reference comparisons have zero material dates but can have missing grids.

Fund	Grid	Modern	Legacy	Time only	Cond. only	B restored
QQQ	15s	0/376	116/376	125/376	3/376	116/376
QQQ	30s	0/376	84/376	81/376	5/376	84/376
QQQ	1m	0/376	46/376	46/376	4/376	46/376
QQQ	5m	0/376	22/376	15/376	7/376	22/376
QQQ	15m	0/376	27/376	5/376	19/376	27/376
QQQ	30m	0/376	29/376	6/376	22/376	29/376
IWM	15s	0/375	164/375	171/375	1/375	14/375
IWM	30s	0/376	99/376	103/376	3/376	11/376
IWM	1m	0/376	61/376	57/376	2/376	6/376
IWM	5m	0/376	17/376	12/376	5/376	6/376
IWM	15m	0/376	26/376	9/376	16/376	21/376
IWM	30m	0/376	30/376	8/376	24/376	26/376
EFA	15s	0/366	142/366	164/366	3/366	110/366
EFA	30s	0/376	114/376	124/376	3/376	70/376
EFA	1m	0/376	59/376	55/376	8/376	36/376
EFA	5m	0/376	35/376	21/376	14/376	30/376
EFA	15m	0/376	42/376	18/376	24/376	37/376
EFA	30m	0/376	76/376	30/376	50/376	69/376
TLT	15s	0/321	65/321	73/321	5/321	34/321
TLT	30s	0/351	45/351	48/351	9/351	24/351
TLT	1m	0/366	32/366	23/366	12/366	23/366
TLT	5m	0/376	32/376	13/376	18/376	28/376
TLT	15m	0/376	47/376	16/376	26/376	40/376
TLT	30m	0/376	53/376	19/376	34/376	49/376

Table 8: Complete A13 RQ primary incidence. Materiality is an absolute log ratio above log(2), with reciprocal lower boundary. The larger threshold and fourth-power units differ from RV; these counts do not estimate a latent-noise distribution.

Fund	Grid	Modern	Legacy	Time only	Cond. only	B restored
QQQ	15s	0/376	113/376	114/376	4/376	113/376
QQQ	30s	0/376	75/376	68/376	4/376	75/376
QQQ	1m	0/376	34/376	34/376	1/376	34/376
QQQ	5m	0/376	13/376	11/376	2/376	13/376
QQQ	15m	0/376	6/376	2/376	2/376	6/376
QQQ	30m	0/376	2/376	0/376	2/376	2/376
IWM	15s	0/375	157/375	151/375	0/375	10/375
IWM	30s	0/376	81/376	89/376	0/376	7/376
IWM	1m	0/376	48/376	51/376	1/376	4/376
IWM	5m	0/376	8/376	10/376	1/376	1/376
IWM	15m	0/376	2/376	4/376	0/376	0/376
IWM	30m	0/376	0/376	1/376	0/376	0/376
EFA	15s	0/366	53/366	61/366	3/366	12/366
EFA	30s	0/376	39/376	37/376	5/376	12/376
EFA	1m	0/376	19/376	17/376	3/376	7/376
EFA	5m	0/376	2/376	1/376	1/376	2/376
EFA	15m	0/376	4/376	2/376	2/376	3/376
EFA	30m	0/376	7/376	2/376	5/376	6/376
TLT	15s	0/321	26/321	17/321	10/321	16/321
TLT	30s	0/351	24/351	15/351	10/351	18/351
TLT	1m	0/366	18/366	10/366	10/366	15/366
TLT	5m	0/376	3/376	1/376	4/376	4/376
TLT	15m	0/376	8/376	2/376	7/376	7/376
TLT	30m	0/376	11/376	1/376	7/376	9/376

Table 9: A13 descriptive H-tilde over lags 1–10 for all fixed constructions/grids. Full 398-session calendar including previously seen January/discovery dates. Missing values retain calendar endpoints, so comparisons may have different endpoint support; the machine-readable H table also gives common-endpoint contrasts and all lag counts. No latent-H confidence interval is implied.

Fund	Grid	Modern	Legacy	Time only	Cond. only	B restored
QQQ	15s	0.1744	0.0668	0.0630	0.1765	0.0668
QQQ	30s	0.1670	0.0795	0.0648	0.1708	0.0795
QQQ	1m	0.1576	0.0931	0.0772	0.1604	0.0931
QQQ	5m	0.1275	0.1037	0.1003	0.1269	0.1037
QQQ	15m	0.0870	0.0825	0.0879	0.0857	0.0825
QQQ	30m	0.0687	0.0685	0.0688	0.0686	0.0685
IWM	15s	0.2007	0.0516	0.0474	0.2033	0.1599
IWM	30s	0.1858	0.0672	0.0623	0.1851	0.1553
IWM	1m	0.1699	0.0915	0.0848	0.1715	0.1565
IWM	5m	0.1258	0.1206	0.1219	0.1271	0.1271
IWM	15m	0.0825	0.0794	0.0830	0.0820	0.0822
IWM	30m	0.0697	0.0686	0.0689	0.0708	0.0703
EFA	15s	0.1063	0.0713	0.0792	0.1093	0.0911
EFA	30s	0.1182	0.1063	0.1022	0.1168	0.1195
EFA	1m	0.1068	0.1035	0.0964	0.1052	0.1105
EFA	5m	0.0935	0.0928	0.0919	0.0929	0.0934
EFA	15m	0.0732	0.0718	0.0720	0.0716	0.0726
EFA	30m	0.0779	0.0779	0.0763	0.0790	0.0799
TLT	15s	0.0605	0.0669	0.0700	0.0602	0.0604
TLT	30s	0.0736	0.0688	0.0734	0.0704	0.0630
TLT	1m	0.0548	0.0621	0.0649	0.0531	0.0557
TLT	5m	0.0453	0.0484	0.0474	0.0468	0.0486
TLT	15m	0.0443	0.0468	0.0451	0.0466	0.0476
TLT	30m	0.0266	0.0289	0.0267	0.0294	0.0296

The complete supplementary tables are `results/etf_breadth_incidence.csv`, `results/etf_breadth_H.csv`, `results/etf_breadth_risk.csv` and `results/etf_breadth_intervals.csv`. They include bands 1–5 and 1–22, the original fixed risk rule for all constructions/grids, and all registered mean-block 22/11/44 intervals. These tables are aggregate research summaries; licensed daily observations are excluded. A13 follows the original result and is not untouched validation. The original A11 policy grid remains explicitly post-result sensitivity; no new preferred policy is selected from this extension.